

Ensemble Methods

TD – PAC-Bayesian Learning – Solutions

M2 Computer Science – MALIA

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Abstract. This tutorial accompanies Sections 2–7 of the lecture notes *PAC-Bayesian Learning: a Theoretical Framework for Ensemble Methods*. The exercises are of increasing difficulty; those marked (★) are at Master level. Notation: $S \sim \mathcal{D}^m$, voters $h : \mathcal{X} \rightarrow \{-1, +1\}$, posterior Q , prior P , Gibbs classifier G_Q , majority vote B_Q , $W_Q(x, y) = \mathbb{E}_{h \sim Q} \mathbb{1}[h(x) \neq y]$, joint error $e_Q = \mathbb{E} W_Q^2$, disagreement d_Q .

Exercise 1 – The Kullback–Leibler divergence

Let $\mathcal{H} = \{h_1, \dots, h_n\}$ be finite and P, Q two distributions on $\mathcal{H}$ with $P_i > 0$.

1. Using Jensen's inequality, show that $\text{KL}(Q\|P) \geq 0$, with equality iff $Q = P$.
2. Assume P uniform. Show that $\text{KL}(Q\|P) = \ln n - H(Q)$ where $H(Q) = -\sum_i Q_i \ln Q_i$. What are the minimal and maximal values of $\text{KL}(Q\|P)$? Compute it when Q is uniform over $k \leq n$ voters.
3. Show that $\text{KL}(Q \otimes Q \| P \otimes P) = 2 \text{KL}(Q\|P)$.
4. Show that $\text{KL}(\mathcal{N}(\mu_1, \sigma^2 I_d) \| \mathcal{N}(\mu_0, \sigma^2 I_d)) = \frac{\|\mu_1 - \mu_0\|^2}{2\sigma^2}$.
5. (★) Binary kl. Fix $q \in [0, 1]$. Show that $p \mapsto \text{kl}(q\|p)$ is convex on $(0, 1)$, minimal at $p = q$, and prove Pinsker's inequality $\text{kl}(q\|p) \geq 2(q - p)^2$ by studying $g(p) = \text{kl}(q\|p) - 2(q - p)^2$.

Solution

1. *Idea:* $-\text{KL}$ is the expectation of a logarithm, and $\ln$ is concave. Let $I = \{i : Q_i > 0\}$ be the support of Q . By definition, $-\text{KL}(Q\|P) = \sum_{i \in I} Q_i \ln \frac{P_i}{Q_i}$. Since the $Q_i, i \in I$, are weights summing to one and $\ln$ is concave, Jensen's inequality gives

$$-\text{KL}(Q\|P) \leq \ln \left(\sum_{i \in I} Q_i \frac{P_i}{Q_i} \right) = \ln \left(\sum_{i \in I} P_i \right) \leq \ln 1 = 0.$$

Equality case. Jensen's inequality for a strictly concave function is an equality iff the variable is constant: $P_i/Q_i = c$ for all $i \in I$. The second inequality is an equality iff $\sum_{i \in I} P_i = 1$, i.e. $I = \mathcal{H}$ (since all $P_i > 0$). Then $c = \sum_i P_i / \sum_i Q_i = 1$ and $Q = P$. Conversely $\text{KL}(P\|P) = 0$.

2. With $P_i = \frac{1}{n}$: $\text{KL}(Q\|P) = \sum_i Q_i \ln(nQ_i) = \ln n \sum_i Q_i + \sum_i Q_i \ln Q_i = \ln n - H(Q)$. The entropy satisfies $H(Q) \geq 0$ (each term $-Q_i \ln Q_i \geq 0$), with equality for a Dirac, and $H(Q) \leq \ln n$ (this is question 1 applied to $\text{KL}(Q\|\text{uniform}) \geq 0$), with equality for the uniform distribution. Hence $0 \leq \text{KL}(Q\|P) \leq \ln n$: the KL is 0 when the posterior is the prior and $\ln n$ when it selects a single voter (the Occam price). For Q uniform on k voters, $H(Q) = k \cdot \frac{1}{k} \ln k = \ln k$ and $\text{KL} = \ln \frac{n}{k}$: selecting a tenth of the voters costs $\ln 10 \approx 2.3$ nats, whatever n .

3. Write $(Q \otimes Q)_{ij} = Q_i Q_j$. Then $\ln \frac{Q_i Q_j}{P_i P_j} = \ln \frac{Q_i}{P_i} + \ln \frac{Q_j}{P_j}$, so

$$\begin{aligned} \text{KL}(Q^2 \| P^2) &= \sum_{i,j} Q_i Q_j \ln \frac{Q_i}{P_i} + \sum_{i,j} Q_i Q_j \ln \frac{Q_j}{P_j} \\ &= \sum_i Q_i \ln \frac{Q_i}{P_i} + \sum_j Q_j \ln \frac{Q_j}{P_j} = 2 \text{KL}(Q \| P), \end{aligned}$$

using $\sum_j Q_j = 1$ in the first sum and $\sum_i Q_i = 1$ in the second. This is why second-order bounds (tandem, disagreement) pay a doubled KL.

4. The two densities have the same normalizing constant, so the log-ratio is $\ln \frac{q(\theta)}{p(\theta)} = \frac{\|\theta - \mu_0\|^2 - \|\theta - \mu_1\|^2}{2\sigma^2}$. Write $\theta = \mu_1 + \sigma\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I_d)$ under Q . Then $\|\theta - \mu_1\|^2 = \sigma^2 \|\varepsilon\|^2$ and $\|\theta - \mu_0\|^2 = \|\mu_1 - \mu_0\|^2 + 2\sigma\langle \varepsilon, \mu_1 - \mu_0 \rangle + \sigma^2 \|\varepsilon\|^2$. The terms in $\|\varepsilon\|^2$ cancel, and $\mathbb{E}\langle \varepsilon, \mu_1 - \mu_0 \rangle = 0$, so $\text{KL} = \mathbb{E}_Q \ln \frac{q}{p} = \frac{\|\mu_1 - \mu_0\|^2}{2\sigma^2}$. Moving the centre of a Gaussian posterior costs the squared distance travelled, measured in units of the variance.
5. For $p \in (0, 1)$, $\partial_p \text{kl}(q \| p) = -\frac{q}{p} + \frac{1-q}{1-p} = \frac{-q(1-p) + p(1-q)}{p(1-p)} = \frac{p-q}{p(1-p)}$, which is negative for $p < q$ and positive for $p > q$: the function decreases then increases, and is minimal at $p = q$ where it vanishes. Moreover $\partial_p^2 \text{kl}(q \| p) = \frac{q}{p^2} + \frac{1-q}{(1-p)^2} > 0$ (unless $q \in \{0, 1\}$, where it is still ≥ 0): the function is convex. Pinsker. Let $g(p) = \text{kl}(q \| p) - 2(q-p)^2$. Then $g(q) = 0$ and

$$g'(p) = \frac{p-q}{p(1-p)} - 4(p-q) = (p-q) \left(\frac{1}{p(1-p)} - 4 \right).$$

Since $p(1-p) \leq \frac{1}{4}$, the bracket is non-negative, so g' has the sign of $p-q$: g decreases on $(0, q)$, increases on $(q, 1)$, and its minimum is $g(q) = 0$. Hence $\text{kl}(q \| p) \geq 2(q-p)^2$. The inequality is tight when $p(1-p) \approx \frac{1}{4}$, i.e. around $p = \frac{1}{2}$, which is why the kl and Hoeffding bounds coincide there.

Exercise 2 – Change of measure and Gibbs posterior

Let $\phi : \mathcal{H} \rightarrow \mathbb{R}$ and P a distribution on a finite set $\mathcal{H}$. Define $Q_\phi(h) = \frac{P(h)e^{\phi(h)}}{\mathbb{E}_P e^\phi}$.

1. Show that for any Q , $\text{KL}(Q \| Q_\phi) = \text{KL}(Q \| P) - \mathbb{E}_Q \phi + \ln \mathbb{E}_P e^\phi$.
2. Deduce the change of measure inequality $\mathbb{E}_Q \phi \leq \text{KL}(Q \| P) + \ln \mathbb{E}_P e^\phi$ and the variational formula $\ln \mathbb{E}_P e^\phi = \max_Q \{\mathbb{E}_Q \phi - \text{KL}(Q \| P)\}$.
3. Let $\lambda > 0$. Show that the minimizer of $Q \mapsto \widehat{R}_S(G_Q) + \frac{1}{\lambda m} \text{KL}(Q \| P)$ is the Gibbs posterior $Q_\lambda(h) \propto P(h)e^{-\lambda m \widehat{R}_S(h)}$, and give the minimal value.
4. Describe Q_λ when $\lambda \rightarrow 0$ and $\lambda \rightarrow +\infty$. Interpret λ .
5. Recover Q_λ directly with a Lagrange multiplier for the constraint $\sum_h Q(h) = 1$.

Solution

1. By definition of Q_ϕ , $\ln \frac{Q(h)}{Q_\phi(h)} = \ln \frac{Q(h)}{P(h)} - \phi(h) + \ln \mathbb{E}_P e^\phi$ (the last term does not depend on h). Taking the expectation under Q : $\text{KL}(Q \| Q_\phi) = \text{KL}(Q \| P) - \mathbb{E}_Q \phi + \ln \mathbb{E}_P e^\phi$.
2. By Exercise 1, $\text{KL}(Q \| Q_\phi) \geq 0$, hence $\mathbb{E}_Q \phi \leq \text{KL}(Q \| P) + \ln \mathbb{E}_P e^\phi$ for every Q : this is the change of measure inequality. It is an equality iff $\text{KL}(Q \| Q_\phi) = 0$, i.e. iff

$Q = Q_\phi$. Therefore $\mathbb{E}_Q \phi - \text{KL}(Q\|P) \leq \ln \mathbb{E}_P e^\phi$ for all Q , with equality at Q_ϕ : the maximum over Q is reached and equals $\ln \mathbb{E}_P e^\phi$. *Interpretation:* the left-hand side of the change of measure concerns a posterior that may depend on the data, the right-hand side only involves the prior; the price of the transfer is exactly the KL.

3. Apply question 2 to $\phi = -\lambda m \hat{R}_S$ (a function of h once S is fixed): $-\lambda m \mathbb{E}_Q \hat{R}_S - \text{KL}(Q\|P) \leq \ln \mathbb{E}_P e^{-\lambda m \hat{R}_S}$, i.e., dividing by $-\lambda m < 0$,

$$\hat{R}_S(G_Q) + \frac{\text{KL}(Q\|P)}{\lambda m} \geq -\frac{1}{\lambda m} \ln \mathbb{E}_{h \sim P} e^{-\lambda m \hat{R}_S(h)},$$

with equality iff $Q = Q_\phi$, i.e. $Q(h) \propto P(h) e^{-\lambda m \hat{R}_S(h)}$. The minimal value is the right-hand side, a “soft minimum” of the empirical risks (it tends to $\min_h \hat{R}_S(h)$ when $\lambda \rightarrow \infty$ on a finite class).

4. When $\lambda \rightarrow 0$, $e^{-\lambda m \hat{R}_S(h)} \rightarrow 1$ for every h and $Q_\lambda \rightarrow P$: the data are ignored. When $\lambda \rightarrow \infty$, divide numerator and denominator by $e^{-\lambda m r^*}$ with $r^* = \min_h \hat{R}_S(h)$: the weight of every h with $\hat{R}_S(h) > r^*$ tends to 0, and Q_λ converges to P restricted to the empirical risk minimizers and renormalized. λ is an inverse temperature: it sets how much a difference of one training error changes the relative weights (by a factor e^λ).
5. The objective $F(Q) = \sum_h Q(h) \hat{R}_S(h) + \frac{1}{\lambda m} \sum_h Q(h) \ln \frac{Q(h)}{P(h)}$ is strictly convex ($x \mapsto x \ln x$ is strictly convex). With the Lagrangian $\mathcal{L}(Q, \nu) = F(Q) + \nu(\sum_h Q(h) - 1)$,

$$\frac{\partial \mathcal{L}}{\partial Q(h)} = \hat{R}_S(h) + \frac{1}{\lambda m} \left(\ln \frac{Q(h)}{P(h)} + 1 \right) + \nu = 0 \iff Q(h) = P(h) e^{-\lambda m \hat{R}_S(h)} e^{-1 - \lambda m \nu}.$$

The constant $e^{-1 - \lambda m \nu}$ is fixed by $\sum_h Q(h) = 1$, which gives the Gibbs posterior. The positivity constraints are automatically satisfied, and by strict convexity this stationary point is the unique minimum.

Exercise 3 – From the general theorem to classical bounds

Recall the general theorem: for convex Δ , w.p. $\geq 1 - \delta$, for all Q , $\Delta(\hat{R}_S(G_Q), R_D(G_Q)) \leq \frac{1}{m} [\text{KL}(Q\|P) + \ln \frac{1}{\delta} \mathbb{E}_{S'} \mathbb{E}_{h \sim P} e^{m \Delta(\hat{R}_{S'}(h), R_D(h))}]$.

1. Explain precisely where each of the following is used in the proof: convexity of Δ , the change of measure, Markov’s inequality, the independence of P from S .
2. (*Hoeffding-type bound*). Take $\Delta(q, p) = \lambda(p - q)$ for a fixed $\lambda > 0$. Using Hoeffding’s lemma, show that w.p. $\geq 1 - \delta$, for all Q : $R_D(G_Q) \leq \hat{R}_S(G_Q) + \frac{\text{KL}(Q\|P) + \ln \frac{1}{\delta}}{\lambda m} + \frac{\lambda}{8}$.
3. Minimize the right-hand side in λ . Why is it not allowed to plug this value into the bound? Propose a correct way to optimize λ and give the resulting price.
4. (*Catoni’s bound*). Let $C > 0$ and $\mathcal{F}_C(p) = -\ln(1 - p(1 - e^{-C}))$. Show that for a fixed voter with risk p and the 0-1 loss, $\mathbb{E}_{S'} e^{-C m \hat{R}_{S'}(h)} = (1 - p(1 - e^{-C}))^m$. Deduce Catoni’s bound.
5. Numerical application: $m = 1000$, $\delta = 0.05$, $\hat{R}_S(G_Q) = 0.1$, $\text{KL}(Q\|P) = 5$. Compute McAllester’s bound $\hat{R}_S + \sqrt{(\text{KL} + \ln \frac{2\sqrt{m}}{\delta})/(2m)}$ and the Hoeffding-type bound for $\lambda \in \{0.1, 0.2\}$.

Solution

1. The proof has three steps (Section 3.2 of the notes). *Convexity of Δ* is used in the first step (Jensen): the risks of G_Q are averages over Q of the risks of the voters, and $\Delta(\mathbb{E}_Q \hat{R}_S(h), \mathbb{E}_Q R_D(h)) \leq \mathbb{E}_Q \Delta(\hat{R}_S(h), R_D(h))$. This reduces the problem to single voters. *The change of measure* is the second step: $\mathbb{E}_Q[m\Delta] \leq \text{KL}(Q\|P) + \ln \mathbb{E}_P e^{m\Delta}$. It removes Q from the random part of the inequality; since it holds for all Q at once, so will the final bound. *Markov's inequality* is the third and only probabilistic step: the non-negative variable $Z(S) = \mathbb{E}_P e^{m\Delta(\hat{R}_S(h), R_D(h))}$ satisfies $Z(S) \leq \mathbb{E}_{S'} Z(S')/\delta$ with probability $\geq 1 - \delta$. *Independence of P and S* is used inside this step, to exchange $\mathbb{E}_{S'}$ and $\mathbb{E}_{h \sim P}$ (Fubini): for each fixed h , $\hat{R}_{S'}(h)$ is an average of i.i.d. variables, whose exponential moment can be computed. If P depended on S' , this exchange would be meaningless.
2. Here $\Delta(q, p) = \lambda(p - q)$ is linear, hence convex. For a fixed h , $\hat{R}_{S'}(h) = \frac{1}{m} \sum_i \ell_i$ with i.i.d. $\ell_i \in [0, 1]$ of mean $R_D(h)$, so by independence

$$\mathbb{E}_{S'} e^{m\lambda(R_D(h) - \hat{R}_{S'}(h))} = \prod_{i=1}^m \mathbb{E} e^{\lambda(R_D(h) - \ell_i)} \leq (e^{\lambda^2/8})^m,$$

by Hoeffding's lemma (a centred variable with range 1 has $\mathbb{E} e^{sX} \leq e^{s^2/8}$). Hence $\mathcal{I}_\Delta(m) \leq e^{m\lambda^2/8}$, and the theorem gives $\lambda(R_D(G_Q) - \hat{R}_S(G_Q)) \leq \frac{1}{m} [\text{KL}(Q\|P) + \ln \frac{1}{\delta}] + \frac{\lambda^2}{8}$. Dividing by λ gives the result.

3. Let $A = \frac{\text{KL}(Q\|P) + \ln(1/\delta)}{m}$. The function $\lambda \mapsto \frac{A}{\lambda} + \frac{\lambda}{8}$ is minimal at $\lambda^* = \sqrt{8A}$, where it equals $\sqrt{A/2}$: we recover a McAllester-type bound $\hat{R}_S(G_Q) + \sqrt{\frac{\text{KL} + \ln(1/\delta)}{2m}}$. *Why it is forbidden*: λ is a parameter of Δ , i.e. of the probabilistic event on which the bound holds; this event must be fixed before drawing S . But λ^* depends on $\text{KL}(Q\|P)$, hence on Q and on S . Using it amounts to choosing, after seeing the data, among infinitely many events, each of probability $\geq 1 - \delta$: nothing guarantees that the chosen one holds. *Correct way*: fix a grid $\lambda_1 < \dots < \lambda_k$ in advance, apply the bound to each λ_j with confidence δ/k (union bound): with probability $\geq 1 - \delta$ all k bounds hold, and we can take the best one. The price is $\frac{\ln k}{\lambda_m}$ (i.e. $\ln k$ added to $\ln \frac{1}{\delta}$) plus the error due to the discretization; a geometric grid with ratio 2 covering $[10^{-3}, 1]$ has $k = 10$ points and costs $\ln 10 \approx 2.3$ nats.
4. For the 0-1 loss, $m\hat{R}_{S'}(h) \sim \text{Bin}(m, p)$ with $p = R_D(h)$, whose moment generating function is $\mathbb{E} e^{sB} = (1 - p + pe^s)^m$. At $s = -C$: $\mathbb{E}_{S'} e^{-Cm\hat{R}_{S'}(h)} = (1 - p + pe^{-C})^m = (1 - p(1 - e^{-C}))^m = e^{-m\mathcal{F}_C(p)}$. Take $\Delta(q, p) = \mathcal{F}_C(p) - Cq$ (convex: $\mathcal{F}_C$ is convex, being $-\ln$ of an affine decreasing function, and $-Cq$ is linear). Then $\mathbb{E}_{S'} e^{m\Delta(\hat{R}_{S'}(h), p)} = e^{m\mathcal{F}_C(p)} e^{-m\mathcal{F}_C(p)} = 1$, so $\mathcal{I}_\Delta(m) = 1$ and, with probability $\geq 1 - \delta$, for all Q : $\mathcal{F}_C(R_D(G_Q)) \leq C\hat{R}_S(G_Q) + \frac{\text{KL}(Q\|P) + \ln(1/\delta)}{m}$. Since $\mathcal{F}_C$ is increasing with inverse $\mathcal{F}_C^{-1}(u) = \frac{1 - e^{-u}}{1 - e^{-C}}$, we obtain $R_D(G_Q) \leq \frac{1}{1 - e^{-C}} [1 - \exp(-C\hat{R}_S(G_Q) - \frac{\text{KL} + \ln(1/\delta)}{m})]$. (For losses in $[0, 1]$ that are not binary, the equality becomes an inequality by convexity of $u \mapsto e^{-Cu}$.)
5. $\ln \frac{2\sqrt{1000}}{0.05} = \ln 1264.9 \approx 7.14$, so McAllester's bound is $0.1 + \sqrt{\frac{5+7.14}{2000}} = 0.1 + 0.078 = 0.178$. For the Hoeffding-type bound, $\ln \frac{1}{0.05} \approx 3.00$ and $\text{KL} + \ln \frac{1}{\delta} = 8.00$. With $\lambda = 0.1$: $0.1 + \frac{8.00}{100} + \frac{0.1}{8} = 0.1 + 0.080 + 0.0125 = 0.1925$. With $\lambda = 0.2$: $0.1 + \frac{8.00}{200} + 0.025 = 0.165$. The optimal value would be $\lambda^* = \sqrt{8 \times 0.008} = 0.253$, giving $0.1 + \sqrt{0.004} = 0.163$. Remember that $\lambda = 0.2$ is legitimate only if it was chosen before seeing the data; if it was selected among $\{0.1, 0.2\}$ after computing both bounds, each must be

computed with $\delta/2$, which gives $0.1 + \frac{5+\ln 40}{200} + 0.025 = 0.1 + 0.0434 + 0.025 \approx 0.168$ for $\lambda = 0.2$.

Exercise 4 – Occam’s razor as a PAC-Bayesian bound

1. Let $\mathcal{H}$ be countable and $Q = \delta_h$ a Dirac on $h \in \mathcal{H}$. Compute $\text{KL}(\delta_h \| P)$. What does Seeger’s bound give for a single voter? Compare with the Occam bound of the lecture notes.
2. Why can we not use Dirac posteriors when $\mathcal{H}$ is uncountable (e.g. linear classifiers with a Gaussian prior)?
3. Let $\mathcal{H} = \{h_1, h_2, \dots\}$ be countable and $P(h_k) = \frac{6}{\pi^2 k^2}$. Check that P is a probability and give the complexity term of h_k . Interpret this prior in terms of *description length*.

Solution

1. $\text{KL}(\delta_h \| P) = \sum_{h'} \delta_h(h') \ln \frac{\delta_h(h')}{P(h')} = 1 \cdot \ln \frac{1}{P(h)}$. Seeger’s bound with $Q = \delta_h$ (for which $\hat{R}_S(G_{\delta_h}) = \hat{R}_S(h)$) gives, simultaneously for all h ,

$$R_{\mathcal{D}}(h) \leq \overline{\text{kl}}^{-1} \left(\hat{R}_S(h), \frac{\ln \frac{1}{P(h)} + \ln \frac{2\sqrt{m}}{\delta}}{m} \right).$$

The Occam bound of the notes has $\ln \frac{1}{\delta}$ instead of $\ln \frac{2\sqrt{m}}{\delta}$: it is slightly tighter because it only covers Dirac posteriors (a union bound over countably many events), whereas the PAC-Bayesian bound covers all posteriors simultaneously. For $m = 1000$ the difference is $\ln(2\sqrt{1000}) \approx 4.1$ nats.

2. If P has a density (e.g. a Gaussian on $\mathbb{R}^d$), then $P(\{h\}) = 0$ for every h , and δ_h is not absolutely continuous with respect to P : $\text{KL}(\delta_h \| P) = +\infty$ and the bound is void. One must use “spread” posteriors, e.g. $\mathcal{N}(\mathbf{w}, \sigma^2 I)$ around the chosen classifier; the bound then certifies the Gibbs classifier around $\mathbf{w}$, and the margin or the majority vote relates it to $\mathbf{w}$ itself (Exercise 6).
3. $\sum_{k \geq 1} \frac{1}{k^2} = \frac{\pi^2}{6}$, so $\sum_k P(h_k) = 1$ and P is a probability. The complexity of h_k is $\ln \frac{1}{P(h_k)} = 2 \ln k + \ln \frac{\pi^2}{6} \approx 2 \ln k + 0.50$. *Description length*: by Kraft’s inequality, any prefix code assigning $\ell(h)$ nats to h defines a sub-probability $P(h) = e^{-\ell(h)}$, and conversely. The prior thus encodes a way of describing hypotheses; the bound says that a hypothesis is cheap if it has a short description *chosen before seeing the data*. Here the index k is described with about $2 \ln k$ nats, so hypotheses listed early are favoured: this is Occam’s razor made quantitative.

Exercise 5 – Bounds on the risk of the majority vote

We assume voters in $\{-1, +1\}$, 0-1 loss, and count ties as errors.

1. Show that $R_{\mathcal{D}}(B_Q) = \mathbb{P}(W_Q \geq \frac{1}{2})$ and $R_{\mathcal{D}}(G_Q) = \mathbb{E}W_Q$. Deduce $R_{\mathcal{D}}(B_Q) \leq 2R_{\mathcal{D}}(G_Q)$. Give an example of distribution of W_Q for which this inequality is an equality.
2. Show that $e_Q = \mathbb{E}[W_Q^2]$, $d_Q = \mathbb{E}[2W_Q(1 - W_Q)]$, and $R_{\mathcal{D}}(G_Q) = e_Q + \frac{1}{2}d_Q$. Show also that $d_Q \leq 2R_{\mathcal{D}}(G_Q)(1 - R_{\mathcal{D}}(G_Q))$.
3. Prove the tandem bound $R_{\mathcal{D}}(B_Q) \leq 4e_Q$. Show that it is better than the first-order bound iff $d_Q \geq R_{\mathcal{D}}(G_Q)$.

4. For $\mu < \frac{1}{2}$, show that $R_{\mathcal{D}}(B_Q) \leq \frac{\mathbb{E}(W_Q - \mu)^2}{(1/2 - \mu)^2}$. Find the optimal μ^* and show that the optimal value is $\frac{\text{Var } W_Q}{\text{Var } W_Q + (1/2 - \mathbb{E} W_Q)^2}$ (Cantelli's inequality).
5. Deduce the C-bound $R_{\mathcal{D}}(B_Q) \leq 1 - \frac{(1 - 2R_{\mathcal{D}}(G_Q))^2}{1 - 2d_Q}$ (when $R_{\mathcal{D}}(G_Q) < \frac{1}{2}$).
6. (*Condorcet*). Q is uniform over n voters erring independently with probability p . Compute $R_{\mathcal{D}}(G_Q)$, e_Q , d_Q as functions of n and p . For $p = 0.3$, compute the exact risk of the vote and the three bounds for $n = 3$, then the C-bound for $n = 25$ and its limit when $n \rightarrow \infty$. Comment.

Solution

1. For a fixed (x, y) , $y \mathbb{E}_Q h(x) = \mathbb{E}_Q[yh(x)] = \mathbb{E}_Q[1 - 2\mathbb{1}[h(x) \neq y]] = 1 - 2W_Q(x, y)$, because $yh(x) = 1$ when h is right and -1 when it errs. The vote errs (ties counted as errors) iff $y \mathbb{E}_Q h(x) \leq 0$, i.e. iff $W_Q(x, y) \geq \frac{1}{2}$. Taking the probability over $(x, y) \sim \mathcal{D}$: $R_{\mathcal{D}}(B_Q) = \mathbb{P}(W_Q \geq \frac{1}{2})$. By Fubini, $\mathbb{E}_{\mathcal{D}} W_Q = \mathbb{E}_Q \mathbb{E}_{\mathcal{D}} \mathbb{1}[h(x) \neq y] = \mathbb{E}_Q R_{\mathcal{D}}(h) = R_{\mathcal{D}}(G_Q)$. Markov's inequality applied to $W_Q \geq 0$ gives $\mathbb{P}(W_Q \geq \frac{1}{2}) \leq \frac{\mathbb{E} W_Q}{1/2} = 2R_{\mathcal{D}}(G_Q)$. *Equality case*: Markov is tight when W_Q only takes the values 0 and $\frac{1}{2}$. Example: two voters with weight $\frac{1}{2}$ each, which never err at the same time, and such that on a fraction r of the points exactly one of them errs. Then $W_Q = \frac{1}{2}$ on these points, the vote errs on them (tie), $R_{\mathcal{D}}(B_Q) = r$ and $R_{\mathcal{D}}(G_Q) = \frac{r}{2}$.
2. For fixed (x, y) , draw $h, h' \sim Q$ independently. The events “ h errs” and “ h' errs” are independent with probability W_Q each, so both err with probability W_Q^2 , and exactly one errs with probability $2W_Q(1 - W_Q)$; for binary voters, h and h' disagree iff exactly one of them errs. Averaging over $\mathcal{D}$ (Fubini): $e_Q = \mathbb{E} W_Q^2$ and $d_Q = \mathbb{E}[2W_Q(1 - W_Q)]$. Pointwise $W_Q = W_Q^2 + \frac{1}{2} \cdot 2W_Q(1 - W_Q)$, hence $R_{\mathcal{D}}(G_Q) = e_Q + \frac{1}{2}d_Q$. Finally Jensen (or $\text{Var } W_Q \geq 0$) gives $e_Q = \mathbb{E} W_Q^2 \geq (\mathbb{E} W_Q)^2 = R_{\mathcal{D}}(G_Q)^2$, so $d_Q = 2(R_{\mathcal{D}}(G_Q) - e_Q) \leq 2R_{\mathcal{D}}(G_Q)(1 - R_{\mathcal{D}}(G_Q))$, with equality iff W_Q is almost surely constant.
3. $\mathbb{P}(W_Q \geq \frac{1}{2}) = \mathbb{P}(W_Q^2 \geq \frac{1}{4}) \leq \frac{\mathbb{E} W_Q^2}{1/4} = 4e_Q$ (Markov applied to W_Q^2). Using question 2, $4e_Q = 4R_{\mathcal{D}}(G_Q) - 2d_Q$, so $4e_Q \leq 2R_{\mathcal{D}}(G_Q) \iff 2R_{\mathcal{D}}(G_Q) \leq 2d_Q \iff d_Q \geq R_{\mathcal{D}}(G_Q)$: the tandem bound improves on the first-order bound exactly when the voters disagree more often than an average voter errs.
4. For $\mu < \frac{1}{2}$ and $W_Q \geq \frac{1}{2}$, we have $W_Q - \mu \geq \frac{1}{2} - \mu > 0$, hence $(W_Q - \mu)^2 \geq (\frac{1}{2} - \mu)^2$: the event $\{W_Q \geq \frac{1}{2}\}$ is included in $\{(W_Q - \mu)^2 \geq (\frac{1}{2} - \mu)^2\}$ and Markov gives the bound. Write $R = \mathbb{E} W$, $e = \mathbb{E} W^2$, $v = e - R^2$ and $f(\mu) = \frac{e - 2\mu R + \mu^2}{(1/2 - \mu)^2}$. Then

$$f'(\mu) = \frac{2(\mu - R)(\frac{1}{2} - \mu) + 2(e - 2\mu R + \mu^2)}{(1/2 - \mu)^3} = \frac{2(e - \frac{R}{2} - \mu(\frac{1}{2} - R))}{(1/2 - \mu)^3},$$

which vanishes at $\mu^* = \frac{R/2 - e}{1/2 - R}$ (when $R < \frac{1}{2}$), is negative before and positive after: μ^* is the minimum. To compute $f(\mu^*)$, note that $\mu^* - R = \frac{R/2 - e - R/2 + R^2}{1/2 - R} = \frac{-v}{1/2 - R}$ and $\frac{1}{2} - \mu^* = \frac{1/4 - R/2 - R/2 + e}{1/2 - R} = \frac{v + (1/2 - R)^2}{1/2 - R}$. The numerator is $e - 2\mu^* R + \mu^{*2} = (\mu^* - R)^2 + v = \frac{v^2}{(1/2 - R)^2} + v = \frac{v(v + (1/2 - R)^2)}{(1/2 - R)^2}$, so

$$f(\mu^*) = \frac{v(v + (1/2 - R)^2)}{(1/2 - R)^2} \cdot \frac{(1/2 - R)^2}{(v + (1/2 - R)^2)^2} = \frac{v}{v + (1/2 - R)^2},$$

which is Cantelli's inequality for W_Q with $a = \frac{1}{2} - R$.

5. With $M = 1 - 2W_Q$ (the margin), $\text{Var } M = 4v$ and $\mathbb{E}M = 1 - 2R$, so $\frac{v}{v+(1/2-R)^2} = \frac{4v}{4v+(1-2R)^2} = \frac{\text{Var } M}{\mathbb{E}M^2} = 1 - \frac{(\mathbb{E}M)^2}{\mathbb{E}M^2}$. Finally $\mathbb{E}M = 1 - 2R_{\mathcal{D}}(G_Q)$ and $\mathbb{E}M^2 = 1 - 4\mathbb{E}W + 4\mathbb{E}W^2 = 1 - 4R_{\mathcal{D}}(G_Q) + 4e_Q = 1 - 2d_Q$ (question 2). Hence $R_{\mathcal{D}}(B_Q) \leq 1 - \frac{(1-2R_{\mathcal{D}}(G_Q))^2}{1-2d_Q}$: the C-bound is the best member of the family of question 4.
6. $nW_Q \sim \text{Bin}(n, p)$ since each voter errs independently with probability p . Hence $R_{\mathcal{D}}(G_Q) = \mathbb{E}W = p$, $e_Q = \mathbb{E}W^2 = \text{Var } W + p^2 = \frac{p(1-p)}{n} + p^2$, and $d_Q = 2(p - e_Q) = 2p(1-p)(1 - \frac{1}{n})$. For $p = 0.3$ and $n = 3$, the vote errs iff at least 2 voters err: $R_{\mathcal{D}}(B_Q) = 3p^2(1-p) + p^3 = 0.189 + 0.027 = 0.216$. The bounds are: first order $2p = 0.6$; $e_Q = 0.09 + 0.07 = 0.16$, so tandem = 0.64; $d_Q = 2 \times 0.21 \times \frac{2}{3} = 0.28$, so C-bound = $1 - \frac{(0.4)^2}{1-0.56} = 1 - \frac{0.16}{0.44} \approx 0.636$. For $n = 25$: $d_Q = 0.42 \times 0.96 = 0.4032$, $1-2d_Q = 0.1936$ and C-bound = $1 - \frac{0.16}{0.1936} \approx 0.174$, while the true risk is about 0.018. When $n \rightarrow \infty$, $1-2d_Q \rightarrow 1-4p(1-p) = (1-2p)^2$ and the C-bound tends to 0, like the true risk, at the rate $\frac{4p(1-p)}{n(1-2p)^2}$. *Comment:* with 3 voters, $d_Q = 0.28 < p = 0.3$, and the second-order bounds are worse than the first-order one (Proposition 5.7 of the notes); with many independent voters, only the second-order bounds, and especially the C-bound, capture the benefit of the vote. The tandem bound tends to $4p^2 = 0.36$ and never goes to 0.

Exercise 6 – Linear classifiers with a Gaussian posterior

Voters $h_{\mathbf{v}}(x) = \text{sign}\langle \mathbf{v}, x \rangle$, prior $P = \mathcal{N}(\mathbf{0}, I_d)$, posterior $Q_{\mathbf{w}} = \mathcal{N}(\mathbf{w}, I_d)$.

1. Show that $B_Q(x) = \text{sign}\langle \mathbf{w}, x \rangle$.
2. Show that $\mathbb{P}_{\mathbf{v} \sim Q_{\mathbf{w}}}(h_{\mathbf{v}}(x) \neq y) = \Phi(-\frac{y\langle \mathbf{w}, x \rangle}{\|x\|})$ where Φ is the standard Gaussian CDF. Give $\hat{R}_S(G_{Q_{\mathbf{w}}})$ and $\text{KL}(Q_{\mathbf{w}}\|P)$.
3. Show that, for a fixed C , minimizing Catoni's bound over $\mathbf{w}$ amounts to minimizing $C \sum_i \Phi(-y_i \langle \mathbf{w}, x_i \rangle / \|x_i\|) + \frac{1}{2} \|\mathbf{w}\|^2$. Compare with the SVM.
4. For $\mathbf{w} = s\mathbf{u}$ with $\|\mathbf{u}\| = 1$, describe the evolution of $\hat{R}_S(G_{Q_{s\mathbf{u}}})$ and of the KL when s grows. Relate to the notion of margin.

Solution

1. Fix x . Under $Q_{\mathbf{w}}$, $\langle \mathbf{v}, x \rangle$ is Gaussian with mean $\langle \mathbf{w}, x \rangle$ and variance $x^\top I_d x = \|x\|^2$. Hence $\mathbb{P}(\langle \mathbf{v}, x \rangle > 0) = \Phi(\frac{\langle \mathbf{w}, x \rangle}{\|x\|})$ and $\mathbb{E}_Q \text{sign}\langle \mathbf{v}, x \rangle = 2\Phi(\frac{\langle \mathbf{w}, x \rangle}{\|x\|}) - 1$, which is positive iff $\langle \mathbf{w}, x \rangle > 0$ (as $\Phi(0) = \frac{1}{2}$ and Φ is increasing). Therefore $B_Q(x) = \text{sign}\langle \mathbf{w}, x \rangle$: the vote of infinitely many noisy linear classifiers is the linear classifier at their centre.
2. Similarly $y\langle \mathbf{v}, x \rangle \sim \mathcal{N}(y\langle \mathbf{w}, x \rangle, \|x\|^2)$ (as $y^2 = 1$), so $\mathbb{P}_{\mathbf{v}}(h_{\mathbf{v}}(x) \neq y) = \mathbb{P}(y\langle \mathbf{v}, x \rangle \leq 0) = \Phi(-\frac{y\langle \mathbf{w}, x \rangle}{\|x\|})$. Averaging over the sample, $\hat{R}_S(G_{Q_{\mathbf{w}}}) = \frac{1}{m} \sum_i \Phi(-\frac{y_i \langle \mathbf{w}, x_i \rangle}{\|x_i\|})$: a smooth *probit* loss of the normalized margin $\frac{y_i \langle \mathbf{w}, x_i \rangle}{\|x_i\|}$. By Exercise 1.4 with $\sigma = 1$, $\text{KL}(Q_{\mathbf{w}}\|P) = \frac{1}{2} \|\mathbf{w}\|^2$.
3. Catoni's bound reads $R_{\mathcal{D}}(G_Q) \leq \mathcal{F}_C^{-1}(C\hat{R}_S(G_Q) + \frac{\text{KL}(Q\|P) + \ln(1/\delta)}{m})$, and $\mathcal{F}_C^{-1}$ is increasing: for fixed C and δ , minimizing the bound over $\mathbf{w}$ amounts to minimizing $C\hat{R}_S(G_{Q_{\mathbf{w}}}) + \frac{\|\mathbf{w}\|^2}{2m}$, i.e. (multiplying by m) $C \sum_i \Phi(-\frac{y_i \langle \mathbf{w}, x_i \rangle}{\|x_i\|}) + \frac{1}{2} \|\mathbf{w}\|^2$. The soft-margin SVM minimizes $C \sum_i \max(0, 1 - y_i \langle \mathbf{w}, x_i \rangle) + \frac{1}{2} \|\mathbf{w}\|^2$: the regularizer is the same (it is the KL to the prior), and the hinge loss is replaced by the probit loss. Two differences: the probit loss is bounded (a badly misclassified point costs at most

1, which gives robustness to label noise) but not convex; and it uses the normalized margin.

4. For $\mathbf{w} = s\mathbf{u}$, the loss of point i is $\Phi(-s\gamma_i)$ with $\gamma_i = \frac{y_i \langle \mathbf{u}, x_i \rangle}{\|x_i\|}$. When s grows, it decreases to 0 for well-classified points ($\gamma_i > 0$), the faster the larger γ_i , and increases to 1 for misclassified points; at $s = 0$ all losses equal $\frac{1}{2}$. Meanwhile the KL grows as $\frac{s^2}{2}$. If all points have a normalized margin at least γ , choosing $s \approx \frac{c}{\gamma}$ makes the empirical Gibbs risk at most $\Phi(-c)$ at a KL cost of $\frac{c^2}{2\gamma^2}$: large margins allow small KL, which is the PAC-Bayesian margin bound (Langford and Shawe-Taylor, 2002). The trade-off in s is the one of Figure 31 of the notes (PBGD).

Exercise 7 – (★) The PAC-Bayes- λ bound and its minimization

1. Prove the refined Pinsker inequality $\text{kl}(q\|p) \geq \frac{(p-q)^2}{2p}$ for $q < p$.
2. Starting from Seeger's bound, show that w.p. $\geq 1 - \delta$, for all Q and all $\lambda \in (0, 2)$ simultaneously, $R_{\mathcal{D}}(G_Q) \leq \frac{\hat{R}_S(G_Q)}{1-\lambda/2} + \frac{\text{KL}(Q\|P) + \ln \frac{2\sqrt{m}}{\delta}}{\lambda(1-\lambda/2)m}$. Why is no union bound over λ needed here, contrary to Exercise 3?
3. For a fixed Q , compute the optimal λ . For a fixed λ , compute the optimal Q on a finite set of voters. Write the alternating minimization algorithm.
4. Explain why the posterior returned by this algorithm on a random forest may give a *worse* majority vote than the uniform one.

Solution

1. Let $0 \leq q < p \leq 1$ and $g(u) = \text{kl}(u\|p) - \frac{(p-u)^2}{2p}$ for $u \in [0, p]$. Then $g(p) = 0$ and $g'(u) = \ln \frac{u(1-p)}{p(1-u)} + \frac{p-u}{p}$. Using $\ln t \leq t - 1$ with $t = \frac{u(1-p)}{p(1-u)}$: $\ln \frac{u(1-p)}{p(1-u)} \leq \frac{u(1-p)-p(1-u)}{p(1-u)} = \frac{u-p}{p(1-u)} \leq \frac{u-p}{p}$, the last step because $u-p < 0$ and $0 < 1-u \leq 1$. Hence $g'(u) \leq 0$: g is non-increasing on $(0, p)$ and $g(q) \geq g(p) = 0$ (see also Appendix A.3 of the notes).
2. Let $\hat{q} = \hat{R}_S(G_Q)$, $p = R_{\mathcal{D}}(G_Q)$ and $\varepsilon = \frac{1}{m}[\text{KL}(Q\|P) + \ln \frac{2\sqrt{m}}{\delta}]$. On the event of Seeger's bound, $\text{kl}(\hat{q}\|p) \leq \varepsilon$. If $p \leq \hat{q}$ the claimed bound is trivial. Otherwise the refined Pinsker inequality gives $\frac{(p-\hat{q})^2}{2p} \leq \varepsilon$, i.e. $p - \hat{q} \leq \sqrt{2p\varepsilon}$. For any $\lambda > 0$, the arithmetic-geometric inequality $\sqrt{ab} \leq \frac{a+b}{2}$ with $a = \lambda p$, $b = \frac{2\varepsilon}{\lambda}$ gives $\sqrt{2p\varepsilon} \leq \frac{\lambda p}{2} + \frac{\varepsilon}{\lambda}$. Hence $p(1 - \frac{\lambda}{2}) \leq \hat{q} + \frac{\varepsilon}{\lambda}$, and for $\lambda \in (0, 2)$ we can divide by $1 - \frac{\lambda}{2} > 0$. *Why no union bound*: all these inequalities are deterministic consequences of the *single* event of Seeger's bound; λ appears only in the algebra, not in the probabilistic statement. In Exercise 3, on the contrary, λ defined the function Δ , hence the event itself.
3. *Optimal λ for fixed Q* . Let $B(\lambda) = \frac{\hat{q}}{1-\lambda/2} + \frac{\varepsilon}{\lambda(1-\lambda/2)}$. Setting $B'(\lambda) = 0$ leads to $\frac{\hat{q}}{2}\lambda^2 = \varepsilon(1-\lambda)$, i.e. $\hat{q}\lambda^2 + 2\varepsilon\lambda - 2\varepsilon = 0$, whose positive root is $\lambda^* = \frac{-\varepsilon + \sqrt{\varepsilon^2 + 2\hat{q}\varepsilon}}{\hat{q}} = \frac{2}{\sqrt{2\hat{q}/\varepsilon + 1} + 1}$. *Optimal Q for fixed λ* . The bound is an increasing affine function of $\hat{R}_S(G_Q) + \frac{\text{KL}(Q\|P)}{\lambda m}$, minimized by the Gibbs posterior $Q(h) \propto P(h)e^{-\lambda m \hat{R}_S(h)}$ (Exercise 2). *Algorithm*: $\lambda \leftarrow 1$; repeat $Q \leftarrow$ Gibbs posterior at λ , then $\lambda \leftarrow \lambda^*(Q)$, until λ stabilizes; return Q and the certificate computed with the kl form. Each step decreases the bound, which is therefore non-increasing along the iterations (and converges to the global minimum under the conditions of Thiemann et al., 2017).

4. The bound only involves the *average* OOB loss $\rho^\top \hat{L}$ and the KL. It therefore puts the weight on the few trees with the smallest OOB loss, which tend to be similar (they are grown on similar bootstrap samples and make similar errors). The effective number of trees collapses (to about 2 on **digits**), the disagreement vanishes, and the vote loses the benefit of averaging: its risk approaches that of a single good tree (0.150 instead of 0.044 on **digits**, Table 2 of the notes). The bound is excellent for the Gibbs classifier, but it is the wrong objective for the vote.

Exercise 8 – (★) Out-of-bag samples

Consider bagging with bootstrap samples of size k drawn with replacement among the m training examples.

1. Compute the probability that a given example is out-of-bag for a given tree; give its limit when $m \rightarrow \infty$ for $k = m$ and for $k = m/2$. Same question for being out-of-bag for two given trees.
2. Explain why the OOB loss $\hat{L}_t$ of tree t is an unbiased estimate of $R_{\mathcal{D}}(h_t)$ given S_t , and why a uniform prior over the *indices* of the trees is a legitimate PAC-Bayesian prior although the trees depend on S .
3. Show that if $\mathbb{E}e^{N \text{kl}(\hat{p}_N \| p)} \leq 2\sqrt{N}$ for all N , then for $n_0 \leq N$, $\mathbb{E}e^{n_0 \text{kl}(\hat{p}_N \| p)} \leq 2\sqrt{n_0}$. Why is this useful?

Solution

1. A bootstrap sample of size k draws k indices uniformly and independently in $\{1, \dots, m\}$. A given example is never drawn with probability $(1 - \frac{1}{m})^k$. Since $(1 - \frac{1}{m})^m \rightarrow e^{-1}$, this tends to $e^{-1} \approx 0.368$ for $k = m$ and to $e^{-1/2} \approx 0.607$ for $k = m/2$. For two trees with independent bootstrap samples, the example is out-of-bag for both with probability $(1 - \frac{1}{m})^{2k} \rightarrow e^{-2} \approx 0.135$ ($k = m$) or $e^{-1} \approx 0.368$ ($k = m/2$). Halving the bootstrap size thus almost triples the size of the pairwise OOB sets used by the tandem bound, at the price of slightly weaker trees.
2. Condition on the bootstrap sample S_t (and on the internal randomness of the tree). Then h_t is a fixed function, and the OOB examples, which were not used to build it, are still i.i.d. from $\mathcal{D}$ and independent of h_t (they are the examples of S whose index was not drawn; given the drawn indices, their values are independent of S_t). Hence $\mathbb{E}[\hat{L}_t | S_t] = R_{\mathcal{D}}(h_t)$, and Maurer's lemma applies to $\hat{L}_t$ conditionally on S_t . The prior is placed on the *indices* $t \in \{1, \dots, n\}$, not on the trees themselves: the uniform distribution on indices is fixed before seeing the data. The dependence of h_t on S only enters the third step of the proof, where, for each index t , the conditional concentration of $\hat{L}_t$ is used.
3. Write $e^{n_0 \text{kl}(\hat{p}_N \| p)} = (e^{N \text{kl}(\hat{p}_N \| p)})^{n_0/N}$. Since $n_0 \leq N$, $x \mapsto x^{n_0/N}$ is concave, and Jensen gives $\mathbb{E}e^{n_0 \text{kl}} \leq (\mathbb{E}e^{N \text{kl}})^{n_0/N} \leq (2\sqrt{N})^{n_0/N} = \exp(n_0 \frac{\ln(2\sqrt{N})}{N})$. The function $N \mapsto \frac{\ln(2\sqrt{N})}{N}$ is decreasing for $N \geq 1$ (its derivative is $\frac{1/2 - \ln(2\sqrt{N})}{N^2} < 0$), so for $N \geq n_0$ this is at most $\exp(n_0 \frac{\ln(2\sqrt{n_0})}{n_0}) = 2\sqrt{n_0}$. *Usefulness:* the OOB sets have different sizes; this lemma allows us to use a single sample size $n_0 = \min_t |T_t|$ in the complexity term while still computing each loss $\hat{L}_t$ on *all* the OOB examples of tree t .

Exercise 9 – (★) Self-bounding learning

Let $\mathcal{B}(Q, S, \delta)$ be a bound such that $\mathbb{P}_S(\forall Q, R_{\mathcal{D}}(Q) \leq \mathcal{B}(Q, S, \delta)) \geq 1 - \delta$.

1. Let Q_S be the posterior returned by any algorithm run on S . Show that $\mathbb{P}_S(R_{\mathcal{D}}(Q_S) \leq \mathcal{B}(Q_S, S, \delta)) \geq 1 - \delta$. Why does this fail for a bound that holds only for a *fixed* predictor (e.g. Hoeffding's inequality)?
2. You build $K = 5$ random forests with tree depths 1, 2, 4, 8 and unlimited, and compute for each of them the OOB tandem certificate. How must you modify the certificates to be allowed to keep the forest with the smallest one? What is the cost in the complexity term?
3. Among the following choices, which are free, which require a union bound, which are forbidden? (a) the temperature λ of a Gibbs posterior; (b) the parameter C of Catoni's bound; (c) the variance of a Gaussian prior chosen on a grid of 10 values; (d) the centre of a Gaussian prior learned on the sample used in the bound; (e) the number of gradient iterations when minimizing the bound.
4. Let $p = \bar{\text{kl}}^{-1}(q, \varepsilon)$. Show that $\frac{\partial p}{\partial \varepsilon} = \frac{p(1-p)}{p-q}$ and $\frac{\partial p}{\partial q} = \frac{p(1-p)}{p-q} \ln \frac{p(1-q)}{q(1-p)}$. Deduce the gradient of the first-order certificate $2\bar{\text{kl}}^{-1}(\rho^\top \hat{L}, \frac{\text{KL}(\rho\|\pi)+c}{n})$ with respect to ρ .
5. Why is the test-set bound usually tighter than a self-bounding certificate for a random forest? Give one advantage of the self-bounding approach.

Solution

1. Let $E = \{S : \forall Q, R_{\mathcal{D}}(Q) \leq \mathcal{B}(Q, S, \delta)\}$, with $\mathbb{P}(E) \geq 1 - \delta$. For $S \in E$ the inequality holds for every posterior, in particular for Q_S , whatever the way it was computed. Hence $\{S : R_{\mathcal{D}}(Q_S) \leq \mathcal{B}(Q_S, S, \delta)\} \supseteq E$ and has probability $\geq 1 - \delta$. For a fixed-predictor bound, there is one event E_h per predictor, each of probability $\geq 1 - \delta$, but their intersection can have a much smaller probability. The learning algorithm, which looks at S , may precisely choose a predictor for which E_h fails: among 1000 random classifiers of risk 0.5 evaluated on 100 examples, the one with the smallest empirical risk violates the naive Hoeffding bound in 99.8% of the runs (Figure 10 of the notes).
2. Keeping the smallest certificate is a data-dependent choice among $K = 5$ bounds. By the union bound (Proposition 6.3 of the notes), if each certificate is computed with confidence $\delta/5$, all five hold simultaneously with probability $\geq 1 - \delta$, and so does the smallest one. For the tandem bound, the complexity term $\frac{2\text{KL} + \ln(2\sqrt{n_0}/\delta)}{n_0}$ becomes $\frac{2\text{KL} + \ln(2\sqrt{n_0}/\delta) + \ln 5}{n_0}$: the price is $\ln 5 \approx 1.6$ nats, i.e. 0.004 in the complexity term for $n_0 = 400$.
3. (a) *Free*: a temperature only indexes a family of posteriors compared with the same prior in the same bound. (b) *Union bound*: C defines the function Δ , hence the event; use a grid fixed in advance with δ/k . (c) *Union bound*: each variance defines a different prior, hence a different bound; cost $\ln 10$. (d) *Forbidden*: the prior would depend on the sample used in the bound, and step 3 of the proof fails; it becomes legitimate if the centre is learned on a separate part S_1 and the bound computed on S_2 (Proposition 6.5). (e) *Free*: any stopping rule produces a posterior, and the bound holds for all posteriors.
4. Let $F(q, p) = \text{kl}(q\|p) - \varepsilon$, so that $F(q, \bar{\text{kl}}^{-1}(q, \varepsilon)) = 0$. We have $F_q = \partial_q \text{kl}(q\|p) = \ln \frac{q(1-p)}{p(1-q)}$ and $F_p = \frac{p-q}{p(1-p)} \neq 0$ (since $p > q$ when $\varepsilon > 0$). By the implicit function theorem, $\frac{\partial p}{\partial \varepsilon} = -\frac{F_\varepsilon}{F_p} = \frac{1}{F_p} = \frac{p(1-p)}{p-q}$ and $\frac{\partial p}{\partial q} = -\frac{F_q}{F_p} = \frac{p(1-p)}{p-q} \ln \frac{p(1-q)}{q(1-p)}$. For the first-

order certificate $B(\rho) = 2 \bar{\text{kl}}^{-1}(q(\rho), \varepsilon(\rho))$ with $q(\rho) = \rho^\top \hat{L}$ and $\varepsilon(\rho) = \frac{\text{KL}(\rho \parallel \pi) + c}{n}$, we have $\nabla_\rho q = \hat{L}$ and $\nabla_\rho \varepsilon = \frac{1}{n}(\ln \frac{\rho}{\pi} + \mathbf{1})$, so by the chain rule

$$\nabla_\rho B = 2 \left[\frac{\partial p}{\partial q} \hat{L} + \frac{\partial p}{\partial \varepsilon} \frac{1}{n} \left(\ln \frac{\rho}{\pi} + \mathbf{1} \right) \right].$$

(With a softmax parametrization $\rho = \text{softmax}(\theta)$, multiply by the Jacobian: $\nabla_\theta B = \rho \odot (g - \rho^\top g)$.) Numerically, at $q = 0.1$, $\varepsilon = 0.05$, $p \approx 0.220$: $\partial_\varepsilon p \approx 1.43$ and $\partial_q p \approx 1.33$.

5. The test-set bound certifies the vote itself, with a single kl inversion, no KL term, and no factor 2 or 4 coming from the Gibbs-to-vote relation; on a random forest this relation is the main source of looseness. In exchange, the test examples are not used for learning, and a new split is needed for model selection. The self-bounding approach uses all the data, provides a learning objective with a guarantee, and allows model selection at the price of $\ln K$; with data-dependent priors it can be competitive when enough data are available.